

Causal & Predictive Analysis of Heat Stress on Punjab Grid Electricity Demand

DAI Mission — Data & AI in Economics | TU Dortmund

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1 — Motivation & Economic Relevance

Why does this matter?

- Punjab is India's agricultural heartland: **irrigation pump sets, cold storage, and agri-processing** create a uniquely weather-elastic grid
- Summer peak demand in 2024 hit **18,441 MW**, a 10-year record, during a concurrent heatwave and monsoon lag
- PSEB operated at a sustained **3,000+ MW deficit** in peak summer 2022 and 2024, forcing rolling load-shedding in rural and urban areas
- Climate projections (IMD 2023) forecast a **15–20% rise** in heatwave frequency over the Indo-Gangetic Plain by 2035

Research question

What is the **marginal causal effect** of continuous heat stress (THI) on hourly electricity demand (MW) in the Punjab power grid, controlling for seasonal, temporal, and environmental confounders and how robustly can supervised and generative architectures **predict and simulate** demand trajectories under extreme heat scenarios?

Three questions, three methods

Why does demand spike?	Causal Inference
How much can we predict?	Supervised Learning
How can we simulate?	Generative

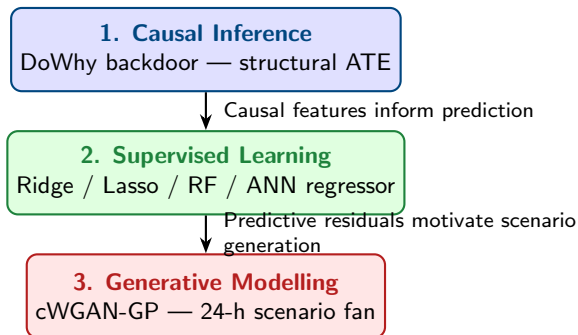
Answering all three gives the Punjab State Electricity Board a complete toolset for **capacity planning, tariff policy, and climate adaptation**.

2 — Data & Methodology

Dataset

- **ICED / NITI Aayog** — hourly Punjab grid load 2016–2025
- **NASA POWER API** — surface meteorology averaged across 23 Punjab districts (hourly)
- **87,672 observations**, THI range 37.1–89.9, demand range 1,793–18,441 MW
- Treatment: $THI = T_F - (0.55 - 0.0055 \cdot RH) \cdot (T_F - 58)$
- Outcome: `demand_mw` (state grid load, MW)
- Confounders: `hour`, `dow`, `month`, `cloud_pct`, `wind_ms`, `precip_mm`

Three-block analytical pipeline



Each block answers a distinct question; together they form a **connected evidence chain** from mechanism to policy action. The causal block identifies *what drives demand*; the supervised block *quantifies future levels*; the generative block *stress-tests grid capacity* under unseen heat extremes.

3 — Results — Causal Inference

Step 1 — Structural Causal Effect

DoWhy backdoor adjustment on **87,672 hourly observations**:

$$\widehat{ATE} = 268.23 \text{ MW/THI}$$

Step 2 — Non-linear Structural Model

Quadratic model ($R^2 = 0.59$):

$$\hat{y} = 23500 - 689.80(\text{THI}) + 6.5321(\text{THI}^2) + \Gamma \mathbf{X}$$

Marginal Causal Effect

$$\text{MCE}(\text{THI}) = -689.80 + 13.0642 \text{ THI}$$

MCE by Operating Scenario

Scenario	THI	MCE
Mild Summer	65	159.37
Average Summer	73.3	268.23
Heatwave	82	381.46

Policy Implication

Heatwave (THI 78→83):

$$\int_{78}^{83} \text{MCE } d(\text{THI}) = 1809.34 \text{ MW}$$

Refutation Tests

Test	Result
Random common cause	Consistent
Placebo treatment	≈ 0
90% subset	Consistent

4 — Results — Prediction & Scenario Generation

From mechanism to prediction

The causal block told us *why* demand rises with THI. The supervised block asks: *how much we can forecast?* THI, hour, month, and weather confounders form the feature set for prediction.

Model	RMSE (MW)	R^2
Ridge / Lasso	2,183	0.531
ANN	1,637	0.736
Random Forest	1,562	0.760

Hold-out: 2024–2025 (17,544 obs)

Random Forest captures non-linear interactions that linear estimators miss, consistent with the accelerating MCE we identified causally. Cross-validation $R^2 = 0.720 \pm 0.041$ across 5 temporal folds confirms robustness beyond the train window.

From prediction to scenario generation

Even the best model gives a *point forecast*. For capacity planning, utilities need *probabilistic scenario fans*. The cWGAN-GP generates 10+ 24-hour demand profiles conditioned on:

- Previous day's actual demand (24 values)
- Previous & current day's weather (24 h $\times$ 5 vars each)

Metric	Result
Wasserstein Distance	870.8 $\pm$ 583.4 MW
Jensen-Shannon Div.	0.0021 $\pm$ 0.0032
JS $\leq$ 0.05 check	✓ Passed
Hold-out days evaluated	306 (2024–2025)

JS divergence < 0.05 confirms that generated profiles are **statistically indistinguishable in shape** from real demand curves, suitable for heatwave stress-testing and reserve margin planning.

5 — Limitations & Synthesis

The connected evidence chain

- 1 **Causal Inference** established the structural mechanism: backdoor adjustment on $\{\text{rh_pct}, \text{temp_c}\}$ yields an Average Causal Effect of **268.23 MW per THI-unit**, robust to three refutation tests (random common cause, placebo, data subset; $p \geq 0.88$ across all three).
- 2 **Supervised Learning** operationalised the causal insight: Random Forest (RMSE 1,562 MW, $R^2 = 0.760$) outperforms the Ridge/Lasso baseline (RMSE 2,183 MW, $R^2 = 0.531$) by 28%; THI, month and hour are the top predictors in the Random Forest framework.
- 3 **Generative Modelling** extended point forecasts to probabilistic scenario fans: JS = **0.0021** (< 0.05 ✓) and Wasserstein 870.8 MW across 306 unseen hold-out days.

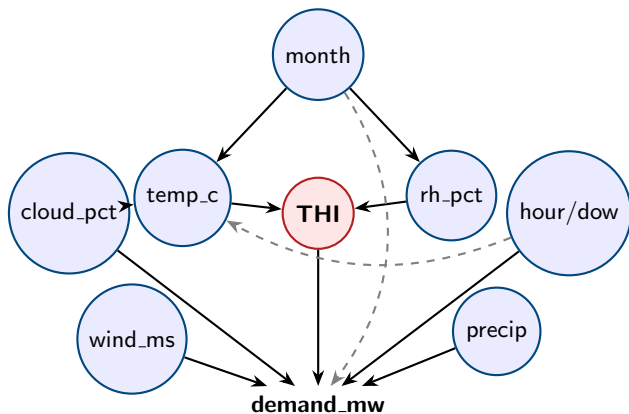
Limitations

- **Unobserved confounders:** unseen factors may inflate the ATE
- **Out-of-distribution tail:** the 2024–2025 record (18,441 MW) lies beyond training range; Random Forest underestimates these peak hours, a known limit of tree-based extrapolation
- **GAN tail gap:** Wasserstein 870.8 MW reflects under-representation of unobserved factors.

References

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Backup — Causal DAG



Backdoor criterion satisfied by adjusting for $\{rh_pct, temp_c\}$ Unconfoundedness assumption: no unmeasured common cause of THI and demand.

Backup — Full Supervised Learning Results

5-Fold Temporal Cross-Validation

Model	RMSE (MW)	MAE (MW)	R^2
Ridge	$2,176 \pm 105$	$1,748 \pm 82$	0.513 ± 0.033
Lasso	$2,175 \pm 105$	$1,748 \pm 81$	0.513 ± 0.032
Random Forest	$1,647 \pm 154$	$1,164 \pm 128$	0.720 ± 0.041
ANN	$1,685 \pm 144$	$1,251 \pm 127$	0.708 ± 0.039

Final Hold-Out Test (2024–2025 — 17,544 observations)

Model	RMSE (MW)	MAE (MW)	R^2
Ridge	2,183.0	1,725.1	0.5312
Lasso	2,182.8	1,725.6	0.5313
ANN	1,637.1	1,165.9	0.7364
Random Forest	1,562.2	1,119.8	0.7599

Best RF hyperparameters: `n_estimators=300`, `max_depth=10`

Train period: 2016–2023 (70,128 obs) Test period: 2024–2025 (17,544 obs)

Backup — cWGAN-GP Architecture & Training

Network Architecture

Component	Input dim	Hidden layers	Output
Generator	32 (noise) + 264 (cond) = 296	512 $\rightarrow$ 1024 $\rightarrow$ 512 (ReLU + BN)	24 (Tanh)
Critic	24 (demand) + 264 (cond) = 288	512 $\rightarrow$ 256 $\rightarrow$ 128 (LeakyReLU 0.2)	1 (unbounded)

Training & Evaluation Summary

Hyperparameter / Metric	Value
Epochs	300 Batch size: 64
Optimiser	Adam ($\eta = 10^{-4}$, $\beta_1 = 0.5$, $\beta_2 = 0.9$)
Critic steps / generator step	5 Gradient penalty $\lambda = 10$
Condition dim	264 (24 lag-demand + 120 lag-weather + 120 curr-weather)
Data augmentation	Gaussian jitter 2–3% σ , $\times 3$ copies
Train days	1,223 (2016-04-02 $\rightarrow$ 2023-08-31), 3,669 after augmentation
Hold-out days	306 (2024-04-01 $\rightarrow$ 2025-08-31), 10 scenarios/day
Wasserstein Distance (MW)	870.8 ± 583.4
Jensen-Shannon Divergence	0.0021 ± 0.0032

Backup — Full Variable Specification

Variable	Type	Role	Description
datetime	Temporal index	Identifier	YYYY-MM-DD HH:00:00
demand_mw	Continuous float	Outcome	Punjab grid consumption (MW)
thi	Continuous float	Treatment	Temperature-Humidity Index
temp_c	Continuous float	Upstream cause	23-station avg. temperature (°C)
rh_pct	Continuous float	Upstream cause	23-station avg. humidity (%)
precip_mm	Continuous float	Covariate	Hourly precipitation (mm)
cloud_pct	Continuous float	Confounder	Cloud cover (%)
wind_ms	Continuous float	Covariate	Wind speed (m/s)
hour	Integer	Fixed effect	Hour of day (0–23)
dow	Integer	Fixed effect	Day of week (0–6)
month	Integer	Fixed effect	Month (1–12)
year	Integer	Fixed effect	Year (2016–2025)

Dataset dimensions: 87,672 hourly obs $\times$ 12 variables **Panel:** 2016–2025 **Scope:** 23 Punjab districts
THI range: 37.1–89.9 **Demand range:** 1,793–18,441 MW